~~Dynamic force Analysis~~

MOD - 2

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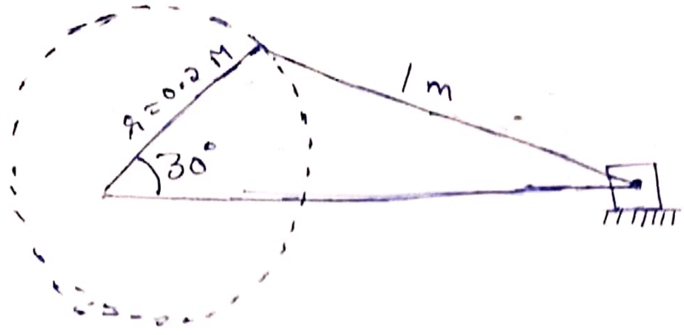
Dynamic Force Analysis

Q) The length of crank and connecting rod of a horizontal engine are 200 mm and 1 m respectively. when the crank is rotating at 400 rpm. when the crank has turned through 30° from the inner dead centre. The difference of pressure b/w cover and piston rod is 0.4 N/mm^2. If the mass of the reciprocating parts is 100 kg and cylinder bore is 0.4 m. Then calculate

(i) Inertia force
(ii) force on piston
(iii) Piston effort
(iv) Thrust on the sides of the cylinder walls
(v) Thrust in the connecting rod, and
(vi) Crank effort (or torque or turning moment)

ans)



$\theta = 30^\circ$

$r = 200$ mm $\Rightarrow$ 0.2 m

$l = 1$ m

$N = 400$ rpm

$P_1 - P_2 = 0.4 \times 10^6$ N/m² (pressure difference)

Mass of reciprocating parts, $m_R = 100$ kg

Mass of cylinder bore, $D = 0.4$ m

$$\omega = \frac{2\pi N}{60}$$

$$= \frac{2\pi \times 400}{60}$$

$\omega = 41.89$ rad/sec

n = Obliquity ratio

$$n = \frac{l}{r} = \frac{100}{0.2} = 5$$

(i) Inertia force, F_1

$$\boxed{F_1 = m_R \omega^2 r \left[\cos\theta + \frac{\cos 2\theta}{n}\right]}$$

$$= 100 \times 41.89 \times 1 \times \left[\cos 30^\circ + \frac{\cos 2 \times 30}{5}\right]$$

$$= 33903.08 \text{ N}$$

(ii) Force on piston, F_L

$$\boxed{F_L = (P_1 - P_2) A}$$

$$A = \frac{\pi}{4} \times D^2$$

$$\Rightarrow 0.4 \cancel{8} \times 10^6 \times \frac{\pi}{4} \times 0.4^2$$

$$F_L = 50265.48 \text{ N}$$

(iii) Piston effect, F_p

$$\boxed{F_p = F_L - F_1}$$

$\Rightarrow$ 33903.08 − 50265.48

$F_p = \underline{16362.4\ N}$

(iv) Thrust on the sides of the cylinder walls

$$\boxed{F_N = F_p \times \tan\phi}$$

where, $\phi = \sin^{-1}\left(\frac{\sin\theta}{n}\right)$

$\phi = 5.73^\circ$

$F_N = F_p \times \tan\phi$

$= 16362.4 \times \tan(5.73^\circ)$

$F_N = \underline{1641.83}$ N

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(v) Thrust in the connecting rod, F_Q ..

$$\boxed{F_Q = \frac{F_P}{\cos\phi}}$$

$$\Rightarrow \frac{16362.4}{\cos(5.73)}$$

$$F_Q = 16444.5 \text{ N}$$

(vi) Crank effort (T),

$$\boxed{T = F_T \times r}$$

where, $F_T = F_Q \sin(\theta + \phi)$

$$= 16444.5 \times \sin(30 + 5.73)$$

$$F_T = 9603.03 \text{ N}$$

$$T = F_T \times r$$

$$= 9603.03 \times 0.2$$

$$T = 1920.6 \text{ Nm}$$

Analysis when weight of connecting rod is given

Q) The connecting rod of a horizontal reciprocating engine is 400 mm and length of the stroke is 200 mm. The mass of the reciprocating parts is 125 kg and that of the connecting rod is 100 kg. The radius of gyration of the connecting rod about an axis ~~connecting~~ through the centre of gravity ~~of the~~ is 120 mm and the ~~rod. distance the~~ distance of centre of gravity of the connectiong rod from big end ~~and~~ centre is 160 mm. The engine runs at 750 rpm. Determine the torque exerted on the crankshaft as the crank has turned 30° from the inner dead centre.

A) Given

$l = \cancel{00}\ 400$ mm $\rightarrow 0.4$ m

~~m_R 0.125 kg~~

$m_R = 125$ kg

$m_c = 100$ kg

$K = 120$ mm $\Rightarrow$ 0.12 m

$N = 750$ rpm

$\omega = \frac{2\pi N}{60} \Rightarrow \frac{2\pi \times 750}{60}$

$\omega = 78.53$ rad/sec

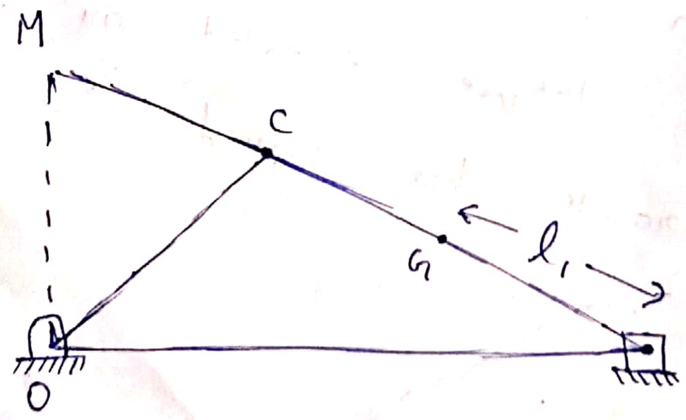
$\theta = 30^\circ$

$G.C = 0.16$ m

$r = \frac{200 \times 10^{-3}}{2} = 0.1$ m

(stroke lengths should be $\div$ by

$n = \frac{l}{r} = \frac{0.4}{0.1} = 4$



Centre of gravity, $\boxed{l_1 = l - G.C}$

$l_1 = 0.4 - 0.16 \Rightarrow l_1 = 0.24$ m

Torque due to inertia force F_1,

$$\boxed{T_1 = F_1 \times OM}$$

$$\boxed{F_1 = \left[m_R + \frac{l - l_1}{l} \times m_c\right] \times \omega^2 \times r \left[\cos\theta + \frac{\cos 2\theta}{n}\right]}$$

$$= \left[125 + \frac{0.4 - 0.24}{0.4} \times 100\right] \times 78.53^2 \times 0.1 \times \left[\cos 30 + \frac{\cos 60}{4}\right]$$

$$= 100841.64 \text{ N} \Rightarrow 100.84 \text{ KN}$$

$$\boxed{OM = r \times \left[\sin\theta + \frac{\sin 2\theta}{2 \times \sqrt{n^2 - \sin^2\theta}}\right]}$$

$$= 0.0609 \text{ m}$$

$$T_1 = F_1 \times OM$$

$$= 100841.64 * 0.0609$$

$$T_1 = 6141.25 \text{ Nm}$$

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Torque due to correction couple,

$$\boxed{T_2 = -m_c l_1 (l - L)\left[\frac{\omega^2 \times \sin(2\theta)}{2n^2}\right]}$$

where,

$$L = \frac{k^2 + l_1^2}{l_1}$$

$$= \frac{0.12^2 + 0.24^2}{0.24}$$

$$L = 0.3$$

$$T_2 = -m_c l_1 (l - L)\left[\frac{\omega^2 \times \sin(2\theta)}{2n^2}\right]$$

$$= -100 \times 0.24 \times (0.4 - 0.3) \times \left[\frac{78.53^2 \times \sin(2 \times 30^\circ)}{2 \times 4^2}\right]$$

$$= -400.66 \text{ N m}$$

Torque exerted due to weight of the connecting rod

$$\boxed{T_3 = {}^{+} m_c \times g \times \frac{l_1}{n} \times \cos\theta}$$

$$= 100 \times 9.81 \times \frac{0.24}{4} \times \cos 30^\circ$$

$$= {}^{+} 50.97 \text{ Nm}$$

So inertia torque on the shaft,

$$T = T_1 - T_2 + T_3$$

$$= 6141.25 - -400.66 + 50.97$$

$$= 6592.5 \text{ Nm}$$

Q) The piston diameter of an internal combustion engine is 125 mm and the stroke is 220 mm. The connecting rod is 4.5 times the crank length and has a mass of 50 kg, The mass of the reciprocating parts is 30 kg. The centre of mass of the connecting rod is 170 mm from the crank pin centre and the radius of gyration of an axis through the centre of mass is 148 mm. The engine runs at 320 rpm. Find the magnitude and the direction of inertia forces and the corresponding torque on the crank shaft when the angle turned by the crank is 140° from the inner dead centre.

Ans) Given

$m_C = 50$ kg

$m_R = 30$ kg

$K = 148$ mm $\Rightarrow 0.148$ m

$N = 320$ rpm

$\theta = 140°$

(stroke) $r = \frac{220}{2} \times 10^{-3} = 0.110$ m

$l = 4.5 \times r$

$l = 4.5 \times 0.11$

$l = 0.495$

(centre of mass) $G.C = 170$ mm $\Rightarrow 0.17$ m

$$\omega = \frac{2\pi N}{60}$$

$$= \frac{2\pi \times 320}{60}$$

$$\omega = 33.51 \text{ rad/sec}$$

$$n = \frac{l}{r} = \frac{0.495}{0.11} \Rightarrow 4.5$$

$$l_1 = l - G.C$$

$$l_1 = 0.325$$

$$D = 125 \text{ mm} \Rightarrow 0.125 \text{ m}$$

$$T_1 = -248.001 \text{ Nm}$$

$$T_2 = 45.568 \text{ Nm}$$

$$T_3 = -27.157 \text{ Nm}$$

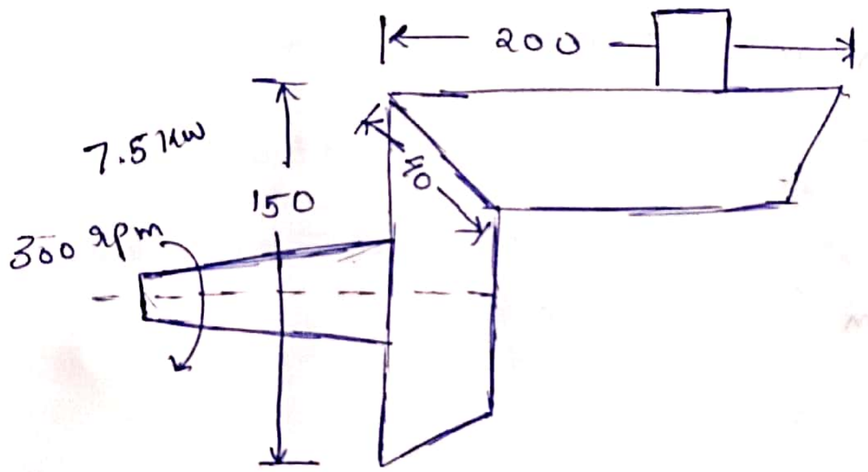
Total inrertia torque,

$$T = T_1 - T_2 + T_3$$

$$T = -320.706 \text{ Nm}$$

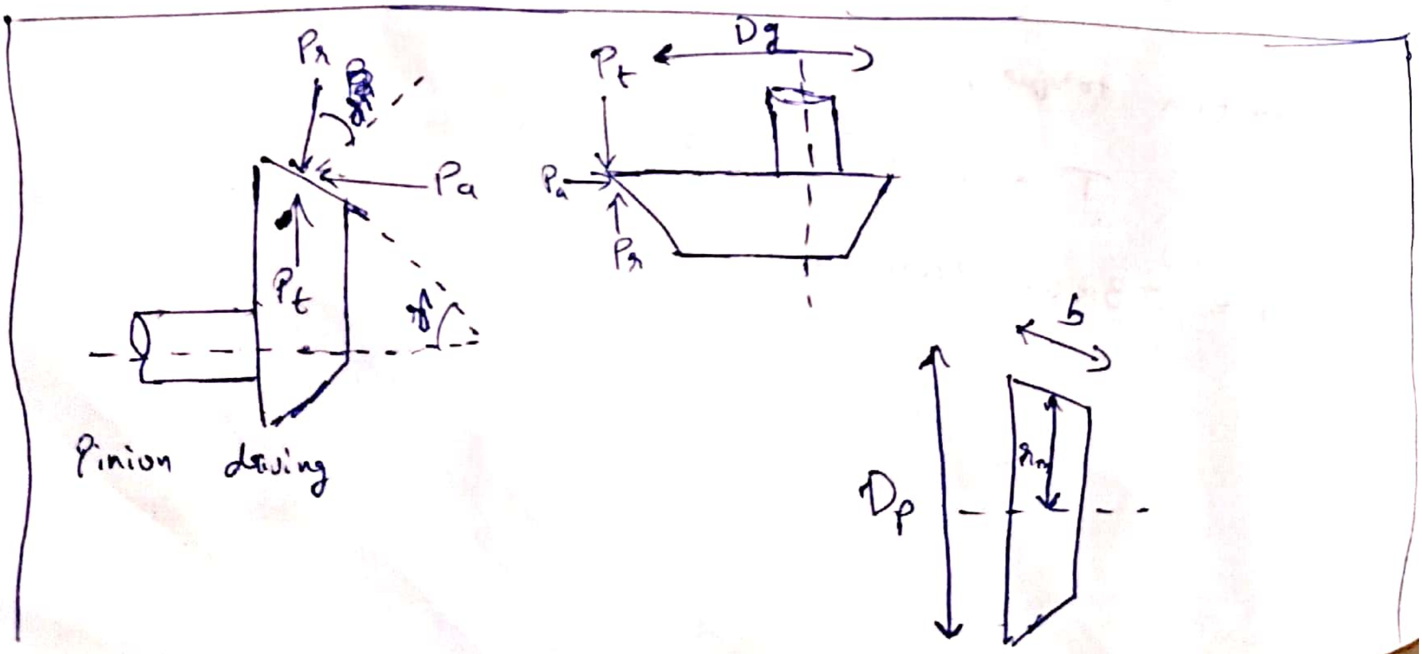
Bevel Gears

Q) A pair of bevel gears trasmitting 7.5 kW at 300 rpm is shown in fig. The pressure angle is 20°. Determine the components of the resultant gear ~~train~~ tooth force and draw a free body diagram of force acting on the piston and the gear.



Given

$P = 7.5$ kW
$N = 300$ rpm
$\alpha = 20°$



ans) Torque,

$$\boxed{T = \frac{60 \times 10^6 \times P \text{ (in kW)}}{2\pi \times N}}$$

P = 7.5 kW

N = 300 rpm

$$T = \frac{60 \times 10^6 \times 7.5}{2\pi \times 300}$$

$$T = 238732.4 \text{ N mm}$$

$$\boxed{\tan \gamma = \frac{Z_p}{Z_g} = \frac{D_p}{D_g}}$$

$$\frac{D_p}{D_g} = \frac{150}{200} \Rightarrow 0.75$$

$$\tan \gamma = 0.75$$

$$\gamma = 36.87^\circ$$

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$$\boxed{r_m = \left[\frac{D_p}{2} - \frac{b \sin \gamma}{2}\right]}$$

$$r_m = \left[\frac{150}{2} - \frac{40 \times \sin(36.87)}{2}\right]$$

$$r_m = 63 \text{ mm}$$

* Tangential force, P_t

$$\boxed{P_t = \frac{T}{r_m}}$$

$$P_t = \frac{238732.4}{63} \Rightarrow 3789.40 \text{ N}$$

* Radial component of force, P_r

$$\boxed{P_r = P_t \times \tan \alpha \times \cos \gamma}$$

$P_r = 3789.40 \times \tan 20^\circ \times \cos 36.87^\circ$

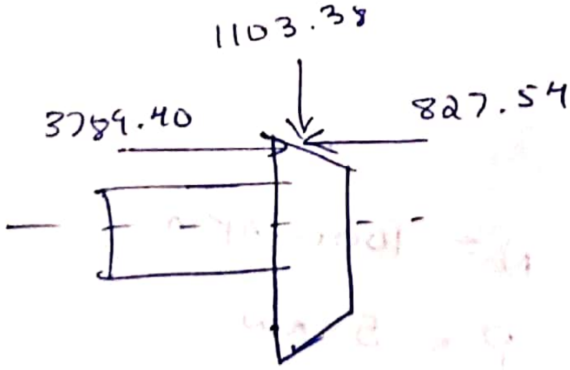
$P_r = \cancel{3779.22}\ 1103.38\ N$

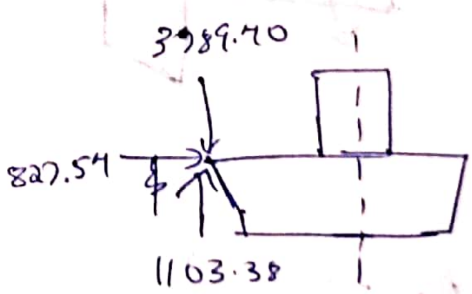
* Thrust component, P_a

$$\boxed{P_a = P_t \times \tan\alpha \times \sin \gamma}$$

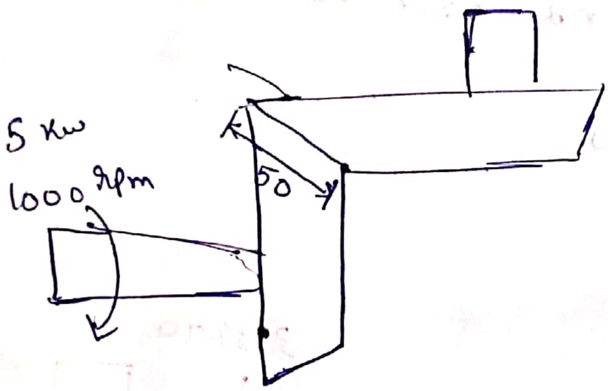
$P_a = 3789.40 \times \tan 20^\circ \times \sin 36.87$

$= \cancel{8}\ 827.54\ N$





Q) Two 20° straight bevel gears have a module pit of 4mm and 24 and 48 teeths respectively. The tooth face width is 50 mm. The pinion rotates at 1000 rpm and trasmits 5 kw. The shafts are at 90°. Determine the components of the gear force and show these on sketch of the gears.



$\alpha = 20^\circ$

$N = 1000$ rpm

$P = 5$ kw

$Z_p = 24$

$b = 50$ mm

$Z_g = 48$

Module, $M = 4$ mm

$\boxed{D_p = m \times Z_p}$

$= 4 \times 24$

$D_p = 96$ mm

$$T = \frac{60 \times 10^6 \times P}{2\pi \times N}$$

$$= \underline{47746.48}$$

$$\tan \gamma = \frac{Z_p}{Z_g}$$

$$\gamma = \tan^{-1}\left(\frac{24}{48}\right)$$

$$\gamma = \underline{26.56^\circ}$$

$$r_m = \frac{D_p}{2} - \frac{b \times \sin \gamma}{2}$$

$$= \frac{96}{2} - \frac{50 \times \sin(26.56)}{2}$$

$$r_m = \underline{36.82} \text{ mm}$$

Tangential force :

$$P_t = \frac{T}{r_m}$$

$$= \frac{47746.48}{36.82}$$

$$P_t = \underline{1296.75} \text{ N}$$

Radial component of force;

$$P_r = P_t \times \tan\alpha \times \cos\gamma$$

$$\Rightarrow 1296.75 \times \tan 20^\circ \times \cos 26.56^\circ$$

$$P_r = 422.16 \text{ N}$$

Thrust component or axial,

$$P_a = P_t \times \tan\alpha \times \cancel{\cos} \sin\gamma$$

$$= 1296.75 \times \tan 20^\circ \times \sin 26.56^\circ$$

$$P_a = 211.03 \text{ N}$$